

Damiano Moscardini

MSc Physics Student at University of Pisa



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Website

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GitHub

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EDUCATION

M.Sc. Physics (Quantum Computing and Technologies)

2025 - 2028 (Expected)

University of Pisa

B.Sc. Physics

2021 - 2025

University of Pisa

High School Diploma (Scientific)

2015 - 2020

Liceo Scientifico Antonio Vallisneri, Lucca

SELECTED PROJECTS

QSVM Encoding Benchmark

2026

Comparative benchmark of 5 quantum encoding strategies against a classical RBF baseline, across 6 datasets with different geometries. Evaluates how well each encoding's kernel geometry matches the classification task, using dedicated metrics (Kernel-Target Alignment, Geometric Coefficient) alongside standard accuracy.

→ <https://github.com/damianomoscardini/qsvm-encoding-benchmark>

3D-Printed Mach-Zehnder Interferometer

2026

Design, construction and optical alignment of a desktop-sized, 3D-printed Mach-Zehnder interferometer to demonstrate wave interference. Fringe intensity acquisition via OPT101 photodiodes and a Siglent SDS824XHD oscilloscope, with thermally induced phase shift via a Kapton heater.

→ <https://github.com/damianomoscardini/3d-printed-mzi>

3D-Printed "Single-Photon" Diffraction Experiment (B.Sc. thesis)

2025

Design and construction of a low-cost, 3D-printed experimental setup to demonstrate diffraction in the "single-photon" regime using an attenuated laser, with the pattern recorded on photographic film as a low-cost "single-photon" detector.

→ <https://github.com/damianomoscardini/3d-printed-single-photon-diffraction>

SKILLS

Programming & Software: Python (PennyLane, PyVisa) • MATLAB • LabVIEW • LaTeX • Typst • AutoCAD

Hardware & Tools: 3D printing • Arduino • Raspberry Pi

Languages: Italian (native) • English (C1 - First Certificate)

EXPERIENCE

Mathematics and Physics Tutor

2020 - Present

Provide continuous academic support for high school and university students. Develop personalized teaching methodologies and preparation strategies for national exams and university-level courses. Collaborated with Centro Yep! (Lucca, 2020 - 2023) and continued with private tutoring.

CONFERENCES

IEEE QCE26

Toronto, 2026

International Conference on Quantum Computing & Engineering | Selected as Student Volunteer

CISF26

Rome, 2026

Italian Conference of Physics Students | Attendee and Oral Presenter

SIPE25

Bari, 2025

Spanish Italian Physics Exchange | Attendee and Oral Presenter

LoT25

Pisa and Florence, 2025

Lights of Tuscany | Attendee and Poster Presenter

INTERESTS

- **3D Printing:** Custom DIY projects, prototyping, lab equipment design and everyday functional prints.
- **Piano:** 11 years of practice.

Last updated: September 18, 2026